Choosing the Promoter-Mining Ensemble

How six DNA models were compared, and why three were deployed

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1 Question and protocol

We compared curated-window classification with restricted genomic ranking to choose the promoter-mining ensemble. Tables are historical records, not new evaluations. Curated AP depends on split, prevalence and weighting; the displayed six-model comparison mixes checkpoints and NTv3's 384-bp context.

The step-10 benchmark covers published TSS with same-contig/strand 81-bp windows expanded by 40 bp at each end. Unmatched starts merge at gaps of at most 81 bp. Recall counts covered catalogue TSS; precision proxy is covered TSS/(covered TSS+unmatched peaks), not ordinary matched-peak precision. Incomplete truth leaves unmatched calls unvalidated. Both benchmark strains supplied silver/negative training windows; their published TSS labels were withheld.

2 Curated classification

Earlier T21_final/plain-0.6 tests have 967/3,318 positives (no-skill AP about 0.29). Evo2 uses T15_xs, with 547/3,053 positives (about 0.18), where deployed CNN-v2 AP is 0.726 rather than 0.848. NTv3's high historical score has unresolved overlap/provenance limits.

Table 1: Historical curated-test average precision (higher is better within a fixed evaluation distribution). Splits, checkpoints, input widths and weighting are not identical across rows; this is checkpoint context, not a controlled six-model ranking.

Model	Historical test average precision
Nucleotide Transformer v3	0.960
ProkBERT	0.884
Nucleotide Transformer	0.854
Fusion CNN	0.848
Simple CNN	0.804
Evo2 probe	0.703

3 Standalone restricted benchmark

Evo2 leads this report-grid comparison. Equal stride does not guarantee the same reverse-strand grid; its standalone and paired deployment values are separate runs.

4 Paired ensemble comparison

Deployment intersects coordinates before computing member ranks, then averages average-tie percentiles. These are not probabilities or unanimous votes; the mine uses no calibrator.

The trio leads at 50% recall on both strains. Paired Evo2 leads at 10% recall in C. autoethanogenum, 87.6% versus 87.0%. Adding CNN-v1 reduced the reported proxy, including 61.6% to 58.3% on C. ljungdahlii at 50% recall. This supports the tested combination, not a universal optimum of three members.

Table 2: Restricted candidate-region, step-10 benchmark precision proxy on two *Clostridium* strains, at 10% and 50% published-TSS recall. Higher means fewer benchmark-unmatched peaks, not fewer proven biological false positives. ✓ marks deployed ensemble members.

		C. ljungdahlii		*C. autoethanogenum*	
Model	Deployed	@10%	@50%	@10%	@50%
Evo2 probe	✓	**71.0%**	**58.0%**	**85.8%**	**71.4%**
ProkBERT	✓	64.4%	47.7%	76.7%	64.6%
Fusion CNN	✓	64.4%	47.2%	81.0%	66.0%
Simple CNN	—	49.0%	40.3%	72.6%	63.0%
Nucleotide Transformer	—	43.9%	36.1%	64.9%	58.7%
Nucleotide Transformer v3	—	35.7%	28.0%	58.5%	48.3%

Table 3: Three-model ensemble versus its members: restricted step-10 benchmark precision proxy at 10% and 50% published-TSS recall, on the deployed shared window grid.

	C. ljungdahlii		*C. autoethanogenum*	
	@10%	@50%	@10%	@50%
Three-model ensemble	**74.4%**	**61.6%**	**87.0%**	**76.6%**
Evo2 probe	69.4%	58.4%	87.6%	72.5%
ProkBERT	64.4%	47.7%	76.7%	64.6%
Fusion CNN	64.4%	47.2%	81.0%	66.0%

5 Deployment and limits

We retain fine-tuned ProkBERT, from-scratch fusion CNN-v2 and the frozen-Evo2 probe. Shared training data does not establish independent errors. Deployed weights retain 888 wrong-assembly B. licheniformis positives. Cluster-disjoint splitting does not remove every dependence.

Original inputs are not committed with this report; authenticated checkpoint recovery and separate local run recovery are documented in docs/MODELS.md. New DNA scans need matching reference/grid and Evo2 runtime. The former constant-only NTv3 metric writer is not executable evaluation evidence. Ranks and motif placements prioritize experiments, not activity, strength or regulation.